

📄 Patient Profile

Patient ID / Name

VPD-2024-001

Age

55

Biological Sex

Male

🕒 Clinical History

Symptom Duration (months)

6

☐ Family history of cancer

☐ Current / former smoker

🔍 Analysis Options

☒ Show signaling pathways

☒ Show therapeutic targets

☒ Show screening protocols



PyMutScan — Virtual Patient Diagnosis

Gene Panel Analysis · Cancer Staging · Risk Management · Clinical Recommendations



Symptom Input



Gene Panel



Diagnosis & Staging



Clinical Report

Primary Diagnosis

Clinical Stage

Risk Level

5-Year Survival

Breast

Stag...

Critical

10–25%

CANCER STAGING



Stage IV

Metastatic — distant organ spread confirmed or highly probable.

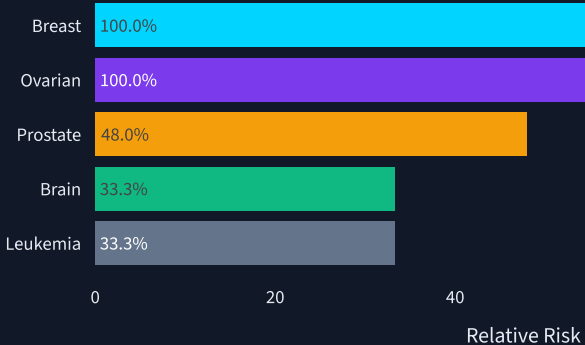
5-YEAR SURVIVAL RATE

10–25%

CLINICAL URGENCY

Critical

Differential Diagnosis — Cancer Probability Ranking



CANCER FORMATION PATHWAY

Breast Cancer — Formation Mechanism:

Oncogenic driver mutations → uncontrolled cell proliferation → evasion of apoptosis → angiogenesis (VEGF upregulation) → local invasion via ECM degradation (MMPs) → lymphovascular spread → distant metastasis via epithelial-mesenchymal transition (EMT).



Mutational Profile

⚠️ TP53

⚠️ BRCA1

✓ EGFR

✓ KRAS

✓ BRCA2

✓ HER2 (ERBB2)

✓ BRAF

✓ PIK3CA

✓ PTEN

✓ RB1

✓ MYC

✓ ALK

✓ VHL

✓ APC

✓ MLH1 / MSH2 (Lynch)

✓ BCL2

✓ CDKN2A (p16)

✓ FGFR2 / FGFR3

 Risk Management Plan — Critical Risk

-  EMERGENCY MULTIDISCIPLINARY BOARD
Same-day MDT review; initiate treatment protocol immediately
-  IMMUNOTHERAPY / PRECISION THERAPY
PD-L1, MSI-H, TMB assessment for checkpoint inhibitor eligibility
-  CLINICAL TRIAL ENROLLMENT
Screen for Phase I/II trials; contact NCI-designated cancer centers
-  PALLIATIVE + CURATIVE DUAL APPROACH
Integrate palliative care from day 1; assess surgical resectability
-  INTENSIVE BIOMARKER MONITORING
Weekly ctDNA + monthly imaging for rapid treatment adaptation
-  HOLISTIC CARE TEAM
Pain management, nutritional, psychological, and spiritual care coordination
-  ADVANCE CARE PLANNING
Discuss prognosis, goals of care, and advance directives with patient/family

 Recommended Screening Protocol — Breast Cancer

- Annual mammography + MRI (high-risk)
- BRCA testing if family history
- Clinical breast exam q6mo